

Decompositions in the Derived Category

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Translator's note. This is a direct English working translation from the project TeX transcription. Mathematical notation and numbering have been kept close to the source; the prose has been normalized into contemporary English.

Let $\mathcal{D}$ be a triangulated category equipped with a t -structure [1, §1.3.1], and let $\mathcal{C}$ be its heart (loc. cit.). We use the notation $\mathcal{D}^{\leq a}$, $\mathcal{D}^{\geq b}$, $\tau_{[a,b]}$, and H^i of [1, §1.3]. Let $K \mapsto K(1)$ be an autoequivalence of $\mathcal{D}$, carrying distinguished triangles to distinguished triangles and respecting the t -structure. We write $K \mapsto K(i)$ for the i -th iterate, $i \in \mathbb{Z}$, of this autoequivalence. We will call the functors $K \mapsto K(i)$ “twists.”

We assume that the objects of $\mathcal{D}$ have bounded cohomology: every object belongs to some $\mathcal{D}^{\leq a}$ and to some $\mathcal{D}^{\geq b}$. If this hypothesis is not satisfied, then the objects considered below should be assumed to have bounded cohomology.

Let K be an object of $\mathcal{D}$, equipped with a morphism

$$\eta: K \rightarrow K(1)[2]. \quad (1)$$

We will again write η for the morphisms deduced from η by twisting (i), shifting, and truncating $\tau_{[a,b]}$, and we write η^i for a composite of i such morphisms. For example, we again write η for the morphisms in the abelian category $\mathcal{C}$

$$\eta: H^i K \rightarrow H^{i+2} K(1) \quad (2)$$

deduced from (1).

Assume that η satisfies the following condition.

(L.V.) For every $i \geq 0$, the i -th iterate of η ,

$$\eta^i: H^{-i}(K) \rightarrow H^i(K)(i),$$

is an isomorphism.

The arguments of [2], recalled in paragraph 1, then show that K is isomorphic to the sum of the $H^i(K)$, each placed in its cohomological degree:

$$K \simeq \bigoplus H^i(K)[-i]. \quad (3)$$

In [2], I announced the existence of a canonical isomorphism (3). The purpose of this note is to define one, and in fact several.

1 Review of the proof of [2]

For $i \geq 0$, the primitive part P^{-i} of $H^{-i}(K)$ is the kernel of

$$\eta^{i+1}: H^{-i}(K) \rightarrow H^{i+2}(K)(i+1).$$

It follows from (L.V.) that the iterated morphisms (2) induce a direct-sum decomposition

$$\sum_{j \geq 0} \eta^j: \bigoplus_{j \geq 0} P^{-i-2j}(-j) \xrightarrow{\sim} H^{-i}(K). \quad (4)$$

For $H^i(K)$, this gives an isomorphism

$$\sum_{j \geq 0} \eta^{i+j} : \bigoplus_{j \geq 0} P^{-i-2j}(-i-j) \xrightarrow{\sim} H^i(K). \quad (5)$$

These isomorphisms assemble into

$$\sum \eta^j : \bigoplus_{i,j} P^{-i}(-j) \xrightarrow{\sim} H^*(K), \quad (6)$$

where the sum is over i, j with $0 \leq j \leq i$. In (6), $P^{-i}(-j)$ is a direct summand of $H^{2j-i}(K)$.

Recall that a cohomological functor F from $\mathcal{D}$ to an abelian category $\mathcal{M}$ is an additive functor such that, for every distinguished triangle $K \rightarrow L \rightarrow M \rightarrow K[1]$, the sequence $F(K) \rightarrow F(L) \rightarrow F(M)$ is exact [6, § 3.1]. Put $F^i(K) = F(K[i])$. By rotating the distinguished triangle $K \rightarrow L \rightarrow M \rightarrow K[1]$, one sees that a cohomological functor F gives a long exact sequence

$$\cdots \rightarrow F^i(K) \rightarrow F^i(L) \rightarrow F^i(M) \rightarrow F^{i+1}(K) \rightarrow F^{i+1}(L) \rightarrow \cdots$$

For K in $\mathcal{D}$, the truncations $\tau_{[a,b]}K$ of K give distinguished triangles

$$\rightarrow \tau_{[a,b]}K \rightarrow \tau_{[a,c]}K \rightarrow \tau_{[b+1,c]}K \rightarrow .$$

The system of corresponding long exact sequences is a “spectral object” in the sense of J. L. Verdier and supplies a spectral sequence

$$E_2^{pq} = F^p H^q K \Rightarrow F^{p+q} K.$$

See the appendix. Let ${}^i E_2^{pq}$ be the analogous spectral sequence for $K(i)$. A morphism (1) gives morphisms of spectral sequences

$${}^i E_2^{pq} \rightarrow {}^{i+1} E_2^{p,q+2}.$$

We show that (L.V.) forces these spectral sequences to degenerate. Proceeding by induction on $r \geq 2$, assume that the d_s vanish for $2 \leq s < r$, and prove that $d_r = 0$. For $q \leq 0$, let ${}^i P^{pq}$ be the direct summand $F^p P^q(i)$ of ${}^i E_2^{pq} = {}^i E_r^{pq}$.

By (4) and (5), it is enough to verify that d_r vanishes on the ${}^i P^{pq}$. By definition, ${}^i P^{pq}$ is annihilated by η^{q+1} . The morphism d_r sends ${}^i E_r^{pq}$ to ${}^i E_r^{p+r,q-r+1}$, on which η^{q+1} is injective. The vanishing of d_r follows.

Applying the degeneration of (6) to a cohomological functor $\text{Hom}(X, K)$, with X in $\mathcal{C}$, one finds that every morphism $X \rightarrow H^i K$ lifts to a degree- i morphism from X to K . In particular, for $X = H^i K$, the identity map of $H^i K$ lifts to a degree- i morphism from $H^i K$ to K . The sum of these morphisms

$$\bigoplus H^i K[-i] \xrightarrow{\sim} K$$

is an isomorphism on cohomology, hence an isomorphism.

2 An isomorphism (3)

Let K, η satisfy (L.V.).

Lemma 1. *Let F be a cohomological functor from $\mathcal{D}$ to the category of abelian groups. Let $i \geq 0$ and let $x \in F(\tau_{\geq -i} K)$ be such that $\eta^{i+1}(x) \in F^{2(i+1)}(\tau_{\geq i+2} K(i+1))$ is zero. Then x has a unique lift y in $F(\tau_{\geq -i-1} K)$ such that $\eta^{i+1}(y) \in F^{2(i+1)}(\tau_{\geq i+1} K(i+1))$ is zero.*

Proof. We saw in paragraph 1 that K is isomorphic to the sum of the $H^i(K)[-i]$. It follows that the sequences

$$0 \rightarrow F^a H^a K \rightarrow F\tau_{\geq a} K \rightarrow F\tau_{\geq a+1} K \rightarrow 0$$

are exact, and similarly for the $K(i)$. The morphism η^{i+1} induces a morphism of short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & F^{i+1} H^{-i-1} K & \longrightarrow & F\tau_{\geq -i-1} K & \longrightarrow & F\tau_{\geq -i} K \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F^{i+1} H^{i+1} K(i+1) & \longrightarrow & F^{2(i+1)} \tau_{\geq i+1} K(i+1) & \longrightarrow & F^{2(i+1)} \tau_{\geq i+2} K(i+1) \longrightarrow 0. \end{array}$$

It remains only to apply the snake lemma. $\square$

The element y of Lemma 1 satisfies the hypothesis of Lemma 1 for $i+1$, and one concludes by induction.

Lemma 2. *Under the hypotheses of Lemma 1, there exists a unique lift $y \in F(K)$ such that, for each $s > i$, $\eta^s y$ has zero image in $F^{2s}(\tau_{\geq s} K(s))$.*

2.3

Apply Lemma 2 to the cohomological functor $F(L) := \text{Hom}(P^{-i}[i], L)$ and to the element $x: P^{-i}[i] \rightarrow \tau_{\geq -i} K$ defined by the inclusion of P^{-i} into $H^{-i} K$. The hypothesis $\eta^{i+1} x = 0: P^{-i}[-i-2] \rightarrow \tau_{\geq i+2} K(i+1)$ is satisfied because $\eta^{i+1}: P^{-i} \rightarrow H^{i+2} K(i+1)$ is zero. We obtain the following.

Proposition 3. *Let $i \geq 0$. There exists a unique morphism f_i from $P^{-i}[i]$ to K such that*

- (i) $H^{-i}(f_i)$ is the inclusion of P^{-i} into $H^{-i} K$;
- (ii) for each $s > i$, the morphism induced by $\eta^s f_i$, from $P^{-i}[i]$ to $(\tau_{\geq s} K)(s)[2s]$, is zero.

2.5

From the morphisms f_i of Proposition 3 and from (6), one deduces morphisms $g_i: H^i(K)[-i] \rightarrow K$ by the rule

$$\text{on the direct summand } P^{-i}(-j) \text{ of } H^{2j-i} \quad (0 \leq j \leq i), \quad g_i \text{ is } \eta^j f_{i-2j}. \quad (7)$$

The sum of these morphisms is an isomorphism (3).

2.6

Let

$$\varphi: \bigoplus H^i(K)[-i] \xrightarrow{\sim} K$$

be an isomorphism inducing the identity on cohomology. Put $Q^{i,j} = P^{-i}(-j)[i-2j]$, so that $P^{-i}(-j)$ lies in cohomological degree $2j-i$. Combining φ with the isomorphism (6), one obtains

$$\varphi_1: \bigoplus_{0 \leq j \leq i} Q^{i,j} \xrightarrow{\sim} K. \quad (8)$$

We denote by $\eta_{i,j}^{k,\ell}$ the matrix entries of $\eta: K \rightarrow K(1)[2]$ relative to this decomposition:

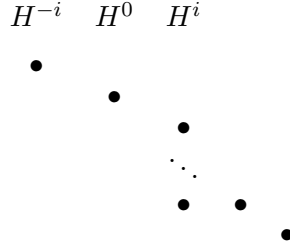
$$\eta_{i,j}^{k,\ell}: Q^{i,j} \rightarrow Q^{k,\ell}(1)[2].$$

These matrix entries are zero for $2j-i < 2\ell-k-2$, since, for A, B in $\mathcal{C}$, one has $\text{Hom}(A[-n], B[-m]) = 0$ for $n < m$. For $2j-i = 2\ell-k-2$, this is the morphism induced by $\eta: H^{2j-i}(K) \rightarrow H^{2j-i+2}(K)(1)$: the identity of $P^{-i}(-j)$ if $k=i$ and $\ell=j+1$, and 0 otherwise.

Proposition 4. *For an isomorphism φ as in 2.6 to be the one defined in 2.5, it is necessary and sufficient that*

- (i) *for $j < i$, the only nonzero $\eta_{i,j}^{k,\ell}$ is the evident identity morphism $\eta_{i,j}^{i,j+1}$;*
- (ii) *$\eta_{i,i}^{k,\ell}$ is nonzero only for $\ell \leq i$.*

The following picture may help one follow the argument. Each $\bullet$ represents a direct summand $Q^{i,j}$. In each row, i is constant. Each column corresponds to one H^n , with $Q^{i,j}$ contributing to H^{2j-i} . Condition (i) determines η , except on the right side of the triangle. Condition (ii) says that $\eta_{i,i}^{k,\ell}$ is nonzero only for k, ℓ in the region to the left of the dotted line.



Notation. We lighten notation by calling an a -morphism from K to L a morphism from K to $L(a)[2a]$. For example, $\eta_{i,j}^{k,\ell}$ is a 1-morphism from $Q^{i,j}$ to $Q^{k,\ell}$.

Proof. Identify K with the sum of the $H^i(K)[-i]$ by means of φ . By definition, the isomorphism φ of 2.5 is characterized by (i) and by the condition

- (*) for every $i \geq 0$ and $s > i$, the s -morphism η^s sends $Q^{i,0}$ into the sum of the $H^n(K)[-n]$ for $n < s$.

Assume (i), so that η is evident on $Q^{i,j}$ for $j < i$. Then condition (*) is equivalent to

- (**) for $t > 0$, the t -morphism η^t sends $Q^{i,i}$ into the sum of the $H^n(K)[-n]$ for $n < i + t$.

*Proof that (**) implies (ii).* We prove by induction on $k - \ell$ that $\eta_{i,i}^{k,\ell} = 0$ if $\ell > i$. Assume that the assertion has been proved for $k - \ell < a$, and prove it for $k - \ell = a$. The restriction of η^{a+1} to $Q^{i,i}$ is the sum

$$\sum \eta^a \circ \eta_{i,i}^{k,\ell}.$$

The terms with $\ell > i$ are controlled by the induction hypothesis and by (i): they are zero if $k - \ell < a$; for $k - \ell \geq a$, they take values in the distinct $Q^{k,\ell+a}$, and the vanishing of $\eta^a \eta_{i,i}^{k,\ell}$ implies the vanishing of $\eta_{i,i}^{k,\ell}$. Note that $Q^{k,\ell+a} \subset H^n[-n]$ for $n = 2\ell + 2a - k$. For $k - \ell = a$, one therefore lands in H^k , and the hypothesis $\ell = k - a > i$ gives $k > i + a$, a value of k forbidden by (**). It remains to show that these terms cannot cancel with those for $\ell \leq i$. It is enough to show that, if $\ell \leq i$, then on $Q^{k,\ell}$ the a -morphism η^a has values in the sum of the $H^n[-n]$ for $n \leq i + a$.

First case: $a \leq k - \ell$. One lands in $H^{-k+2\ell+2a}$, and

$$-k + 2\ell + 2a = (-k + \ell + a) + \ell + a \leq 0 + i + a = i + a.$$

Second case: $a > k - \ell$. Write $\eta^a = \eta^{a-(k-\ell)}\eta^{k-\ell}$, and consider $\eta^{a-(k-\ell)}$ on $Q^{k,k}$. Hypothesis (*) ensures that one lands in the H^n with

$$n < k + (a - (k - \ell)) = a + \ell \leq a + i.$$

Proof that (ii) implies (i). We prove by induction on $a \geq 1$ that the a -morphism η^a restricted to $Q^{i,i}$ has values in the $Q^{k,\ell}$ with $\ell \leq i + a - 1$. For $a = 1$ this is hypothesis (ii). Assume the

claim for a , and consider $\eta^{a+1} = \eta \circ \eta^a$. It is enough to see that, if $\ell \leq i + a - 1$, the restriction of the 1-morphism η to $Q^{k,\ell}$ has values in the sum of the $Q^{m,n}$ with $n \leq i + a$. If $\ell < k$, this follows from (i). If $\ell = k$, it follows from (ii).

Finally $Q^{k,\ell}$ lies in $H^n[-n]$ for $n = 2\ell - k$, and if $\ell \leq i + a - 1$ then

$$n = 2\ell - k = \ell - (k - \ell) \leq \ell \leq i + a - 1.$$

□

3 Another isomorphism (3)

3.1

The opposite category $\mathcal{D}^0$ of $\mathcal{D}$ is again a triangulated category. Its translation functor is $K^0 \mapsto (K[-1])^0$. We equip it with the autoequivalence (1) for which $K^0(1) = K(-1)^0$.

The t -structure of $\mathcal{D}$ supplies one on $\mathcal{D}^0$, and $\eta: K \rightarrow K(1)[2]$ transposes to

$${}^t\eta: K^0 \rightarrow K^0(1)[2].$$

For η to satisfy (L.V.) it is necessary and sufficient that ${}^t\eta$ satisfy (L.V.).

The construction (2.5) is not self-dual, as is seen from its characterization in Proposition 4. The dual construction, obtained from Proposition 3 applied to ${}^t\eta$, gives morphisms

$$f'_i: K \rightarrow P^{-i}(-i)[-i]$$

such that $H^i(f'_i)$ is the projection of $H^i(K)$ onto $P^{-i}(-i)$, and such that, for each $s > i$, the morphism induced by $f'_i \eta^s$, from K to $P^{-i}(-i)[-i](s)[2s]$, is zero on $\tau_{\leq -s}K$. Proceeding as in 2.5, one deduces morphisms $g'_i: K \rightarrow H^i(K)[-i]$ whose sum is an isomorphism from K to the sum of the $H^i(K)[-i]$. This isomorphism is in general not the inverse of the one constructed in 2.5.

In this paragraph we assume that we are in characteristic zero, that is, that the category $\mathcal{D}$ is $\mathbf{Q}$ -linear. Let K, η satisfy (L.V.). Our aim is to construct a self-dual isomorphism (3).

3.2

The grading of H^*K defines an endomorphism

$$h: H^*K \rightarrow H^*K : \text{multiplication by } n \text{ on } H^n K.$$

Passing to cohomology, η gives a morphism of degree 2

$$e: H^*K \rightarrow H^*K(1).$$

Because we are in characteristic zero, one can complete h and e by a morphism of degree -2

$$f: H^*K \rightarrow H^*K(-1)$$

such that h, e, f satisfy the relations between the standard generators of $\mathfrak{sl}(2)$:

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

To give meaning to these formulae, a morphism and those obtained from it by twisting (n) are denoted in the same way.

In characteristic zero, the existence of f is equivalent to (L.V.). In the decomposition (6), each sum over j , $\bigoplus_j P^{-i}(j)$, is stable under h, e, f , and from the factor in H^{i+1} to the factor in H^{i-1} , f is the inverse of e , multiplied by

$$\frac{(i+1)^2}{4} - \frac{n^2}{4}.$$

Let F be one of the bifunctors $\text{Ext}^a(A, B) = \text{Hom}_{\mathcal{D}}(A, B[a])$ on $\mathcal{C}$. Up to canonical isomorphism, the functor $F(A(i), B(j))$ depends only on $n := j - i$. We denote it by $F_n(A, B)$. From h, e, f one again obtains an action of $\mathfrak{sl}(2)$ on $F_*(H^*K, H^*K)$. More precisely, there are morphisms

$$\begin{aligned} h &: F_j(H^*K, H^*K) \rightarrow F_j(H^*K, H^*K), \\ e &: F_j(H^*K, H^*K) \rightarrow F_{j+1}(H^*K, H^*K), \\ f &: F_j(H^*K, H^*K) \rightarrow F_{j-1}(H^*K, H^*K) \end{aligned}$$

satisfying the $\mathfrak{sl}(2)$ relations. The endomorphism h is multiplication by n in degree n , and e , of degree 2, is the bracket with $\eta: H^*K \rightarrow H^*K(1)$.

Let F_j^d be the degree- d part of $F_j(H^*K, H^*K)$: the sum of the $F_j(H^pK, H^qK)$ for $q - p = d$. If $H^iK = 0$ for $|i| > N$, then $F_j^d = 0$ for $|d| > 2N$. On the sum of $\mathbf{Q}$ -vector spaces $\bigoplus_d F_{j+d/2}^d$, where the sum is over d of fixed parity, with j integral for d even and half-integral for d odd, (h, e, f) defines an action of $\mathfrak{sl}(2)$, and e therefore satisfies (L.V.). Reading the corresponding decomposition (6), one obtains the following.

Lemma 5. *With the preceding notation, for $d \leq 1$:*

- (i) $e: F_{j-1}^{d-2} \rightarrow F_j^d$ is injective;
- (ii) F_j^d is the direct sum of the image of e and the kernel of $e^{1-d}: F_j^d \rightarrow F_{j+1-d}^{2-d}$.

3.4

Let φ be an isomorphism

$$\bigoplus H^i(K)[-i] \xrightarrow{\sim} K \quad (9)$$

inducing the identity on cohomology. As in Proposition 4, and using its notation, regard η as a 1-map from K to K , and let $\eta^{(d)}$ be its homogeneous component of degree d in the decomposition (9). With the notation of 3.2, $\eta^{(d)}$ belongs to F_1^d , for F the functor Ext^{2-d} . For $d > 2$, it is therefore zero; for $d = 2$, $\eta^{(2)} \in \text{Hom}(H^*K, H^*K(1))$ is the endomorphism e of 3.2.

Proposition 6. *Assume that $\mathcal{D}$ is $\mathbf{Q}$ -linear and that (K, η) satisfies (L.V.). Then there exists a unique isomorphism (9) for which*

$$\text{for } d \leq 1, \quad e^{1-d}(\eta^{(d)}) = 0. \quad (10)$$

For $d = 1$, condition (10) means that $\eta^{(1)} = 0$.

Proof. By induction on $n \geq -1$, we show that the conditions (10) for $-n < d \leq 1$ determine φ uniquely modulo composition with an endomorphism $1 + w$ of the sum of the $H^i(K)[-i]$, with w of degree $\leq -n - 2$. Note that, with the notation of 3.2, the homogeneous component $\psi^{(d)}$ of degree d of w belongs to F_0^d for F the functor Ext^{-d} .

For $n = -1$, the hypothesis is empty and the conclusion says that φ , assumed to be the identity on cohomology, is defined modulo composition with $1 + \psi$, where ψ has degree ≤ -1 .

We prove the assertion for $n \geq 0$, assuming it known for $n - 1$. Changing φ to $\varphi \circ (1 + \psi)$ conjugates the 1-morphism η – more precisely $\varphi^{-1}\eta\varphi$ – of the sum of the $H^i K[-i]$ to $(1 + \psi)^{-1}\eta(1 + \psi)$. For ψ of degree $\leq -n - 1$, since η has degree ≤ 2 , one has

$$\begin{aligned} (1 + \psi)^{-1}\eta(1 + \psi) &\equiv \eta + [\eta, \psi] \\ &\equiv \eta + e(\psi^{(-n-1)}) \end{aligned}$$

modulo degree $\leq -n$. By Lemma 5, there is a unique choice of $\psi^{(-n-1)}$ giving a new η satisfying (10) for $d > -n$, as required. $\square$

Proposition 7. *Let φ be the isomorphism (9) of Proposition 6. Then, on $P^{-i}[i]$, φ coincides with the morphism f_i of Proposition 3.*

Proof. Identify the sum of the $H^i[-i]$ with K by φ . By the characterization in Proposition 3 of f_i , it is enough to verify that, for $s > i$, the restriction to $P^{-i}[i]$ of the s -morphism η^s has degree $< s + i$. We have a decomposition into homogeneous components

$$\eta = \eta^{(2)} + \sum_{a \leq 0} \eta^{(a)}.$$

Put $e = \eta^{(2)}$. By Proposition 6,

$$(\text{ad } e)^{-a+1}(\eta^{(a)}) = 0.$$

Expand η^s and reorder the factors by repeated application of the rule

$$ex = xe + \text{ad } e(x).$$

One finds that η^s is a sum of terms of the following types:

- (a) e^s ;
- (b) $(\text{ad } e)^{r_k}(\eta^{(a_k)}) \cdots (\text{ad } e)^{r_1}(\eta^{(a_1)})e^{r_0}$ with $\sum_0^k r_k < s$.

If $s > i$, e^s vanishes on $P^{-i}[i]$. Similarly, the terms (b) with $r_0 > i$ vanish on $P^{-i}[i]$. The factor $(\text{ad } e)^{r_k}(\eta^{(a_k)})$ has degree $2r_k + a_k$ and is zero if $r_k > -a_k$. If it is nonzero, its degree is

$$2r_k + a_k = r_k + (r_k + a_k) \leq r_k.$$

Each term of type (b) nonzero on $P^{-i}[i]$ is therefore of degree at most

$$2r_0 + \sum_1^k r_k = r_0 + \sum_0^k r_i < i + s,$$

as required. $\square$

4 ℓ -adic cohomology

4.1

Let k be a field and let ℓ be a prime number different from the characteristic. Take for $\mathcal{D}$ the ℓ -adic derived category with its natural t -structure [5]. The heart $\mathcal{C}$ of $\mathcal{D}$ is the category of ℓ -adic representations (V, ρ) of $\text{Gal}(\bar{k}/k)$: V is finite-dimensional over $\mathbf{Q}$, and $\rho: \text{Gal}(\bar{k}/k) \rightarrow \text{GL}(V)$ is continuous. The $\text{Ext}^a(V, W)$ are the groups $H^a(\text{Gal}(\bar{k}/k), \text{Hom}(V, W))$ computed in terms of continuous cochains.

Let X be projective and smooth over k , purely of dimension N . Let $a: X \rightarrow \text{Spec}(k)$ be the structural morphism. Then $Ra_*\mathbf{Q}_\ell$ is an object of $\mathcal{D}$. Fix an ample invertible sheaf $\mathcal{O}(1)$,

and let $\eta: Ra_*\mathbf{Q}_\ell \rightarrow Ra_*\mathbf{Q}_\ell(1)[2]$ be cup product with $c_1\mathcal{O}(1)$. It is known that $(Ra_*\mathbf{Q}_\ell[N], \eta)$ satisfies (L.V.) (P. Deligne, [4, Theorem 4.1.1]).

The decomposition 2.5 of $Ra_*\mathbf{Q}_\ell[N]$ gives, by translation, a decomposition of $Ra_*\mathbf{Q}_\ell$. Changing notation, we write P^i for the direct summand of $H^i Ra_*\mathbf{Q}_\ell$ given by the kernel of

$$\eta^{N-i+1}: H^i Ra_*\mathbf{Q}_\ell \rightarrow H^{2N-i+2} Ra_*\mathbf{Q}_\ell.$$

In the notation of §1 for $Ra_*\mathbf{Q}_\ell[N]$, this is P^{i-N} . The decomposition (8) deduced from 2.5 translates into a decomposition

$$\bigoplus_{0 \leq j \leq N-i} P^i(-j)[-i-2j] \xrightarrow{\sim} Ra_*\mathbf{Q}_\ell. \quad (11)$$

According to Proposition 4, the only interesting components of the 1-morphism η in this decomposition are the $\eta_{i,N-i}^{k,\ell}$ for $\ell \leq N-i$. One has

$$\eta_{i,N-i}^{k,\ell} \in \text{Ext}^{(2N-i)-(k+2\ell)+2}(P^i(-(N-i)), P^k(-\ell)(1)).$$

One would like to regard these classes as motivic: they have been defined by a procedure uniform in ℓ . On the other hand, optimistic conjectures on the relation between algebraic cycles and the motivic derived category imply [3, §§3.7, 3.8] that, for M an effective motive of weight w and $i \geq 0$, one has

$$\text{Ext}^n(1, \check{M}(i)) = 0 \quad \text{for } n > w + i.$$

The special case which matters to us is the vanishing of this Ext when n is equal to the negative of the weight of $\check{M}(i)$, with M effective and $i > 0$.

In the case at hand, by (L.V.) and Poincaré duality, $\check{P}^k$ is isomorphic to $\check{P}^k(-k)$ and

$$\eta_{i,N-i}^{k,\ell} \in \text{Ext}^n(1, V),$$

where n is the negative of the weight of V , and

$$V = \check{P}^i \otimes \check{P}^k(N-i-k-\ell+1).$$

If the ℓ -adic classes $\eta_{i,N-i}^{k,\ell}$ are motivic, the conjecture therefore implies a positive answer to the following question.

Question. Does one have $\eta_{i,N-i}^{k,\ell} = 0$ for $N-i-k-\ell \geq 0$?

4.2

In the decomposition (11), the cup product

$$Ra_*\mathbf{Q}_\ell \otimes Ra_*\mathbf{Q}_\ell \rightarrow Ra_*\mathbf{Q}_\ell$$

has components

$$c_{i,j;k,\ell}^{m,n} \in \text{Ext}^A(P^i(-j) \otimes P^k(-\ell), P^m(-n)),$$

with A the negative of the weight of V and

$$V = \check{P}^i \otimes \check{P}^k \otimes \check{P}^m(j+k-m-n).$$

Thus one wants vanishing for $j+k-m-n > 0$.

4.3

Let Z be an algebraic cycle of codimension d . It has a class

$$cl(Z): \mathbf{Q}_\ell \rightarrow Ra_* \mathbf{Q}_\ell(d)[2d],$$

with components

$$cl(Z)^{i,j} \in \text{Ext}^A(1, V),$$

where A is the negative of the weight of V and

$$V = \check{P}^i(-j+d) = \check{P}^i(d-i-j).$$

One therefore expects the vanishing of the components $cl(Z)^{i,j}$ for $i+j < d$.

Appendix. Spectral objects, after J. L. Verdier

I learned the formalism presented in this appendix from J. L. Verdier, who said that he was inspired by the treatment of spectral sequences in Cartan–Eilenberg (vol. 1, pp. 318–319).

A.1

Let $\mathcal{D}$ be a triangulated category. A *spectral object* in $\mathcal{D}$ consists of

- (1) a family X_{pq} of objects of $\mathcal{D}$, indexed by pairs of integers $p \leq q$;
- (2) for $p' \leq p$ and $q' \leq q$, a morphism $X_{pq} \rightarrow X_{p'q'}$;
- (3) for $p \leq q \leq r$, a degree-one morphism, called a boundary, $X_{qr} \rightarrow X_{pq}$.

One requires that

- (a) the morphisms (2) define a contravariant functor from the ordered set of pairs (p, q) with $p \leq q$ to $\mathcal{D}$;
- (b) for $p \leq q \leq r$, $p' \leq q' \leq r'$, and $p' \leq p$, $q' \leq q$, $r' \leq r$, the diagram of the morphisms (2) and (3) is commutative;
- (c) for $p \leq q \leq r$, the triangle

$$X_{qr} \rightarrow X_{pr} \rightarrow X_{pq} \xrightarrow{+1} X_{qr}$$

formed by the morphisms (2) and (3) is distinguished.

Example 8. For $\mathcal{D}$ the derived category of an abelian category $\mathcal{A}$ and (K, F) a filtered complex, the objects $X_{pq} = F^p(K)/F^q(K)$ form a spectral object.

Remark 9. A spectral object X is said to have amplitude in $[a, b]$ if the morphisms (2), $X_{pq} \rightarrow X_{p'q'}$, are isomorphisms for $p' \leq p \leq a$ and $q' = q$, and also for $b < q' \leq q$ and $p' = p$. This is equivalent to requiring $X_{p,p+1} = 0$ for $p \notin [a, b]$. A spectral object of amplitude $[0, n]$ with $n = 1$, respectively 2, is identified with a distinguished triangle, respectively an octahedral diagram; see [1, §§ 1.1.6, 1.1.7, 1.1.14].

A.2

Let $\mathcal{A}$ be an abelian category equipped with a “translation functor” $X \mapsto X[1]$, an autoequivalence. Write $X \mapsto X[n]$ for the n -th iterate of the translation functor, and call a morphism of degree n from X to Y a morphism from X to $Y[n]$. A sequence $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z$, with α of degree n and β of degree m , is said to be exact if the sequence $X \rightarrow Y[n] \rightarrow Z[n+m]$ is exact.

A *spectral object* of $\mathcal{A}$ consists of data $(1')$, $(2')$, $(3')$ of objects and morphisms of $\mathcal{A}$, as in A.1, satisfying conditions (a') and (b') with the same formulation as A.1(a),(b), and the condition

(c') for $p \leq q \leq r$, the sequence

$$\cdots \rightarrow X_{qr} \rightarrow X_{pr} \rightarrow X_{pq} \xrightarrow{\partial} X_{qr} \rightarrow X_{pr} \rightarrow \cdots$$

formed by the morphisms $(2')$ and $(3')$ is exact.

Let X be a spectral object of $\mathcal{A}$. From (b') , the morphism $(2')$: $X_{p,r} \rightarrow X_{p,r}$ is the identity. From (c') , for $p = q$, one has $X_{pp} = 0$. If $p \leq q'$, the morphism $X_{pq} \rightarrow X_{p'q'}$ is zero, since it factors through X_{pp} . For $p \leq q \leq r \leq s$, the composite of boundary morphisms

$$X_{pq} \rightarrow X_{qr} \rightarrow X_{rs}$$

is zero: the first morphism factors through the boundary morphism to X_{qs} , followed by $X_{qs} \rightarrow X_{qr}$, and the composite $X_{qs} \rightarrow X_{qr} \rightarrow X_{rs}$ is zero by (c') .

If $\mathcal{A}$ is the category of graded objects of an abelian category $\mathcal{B}$, with the translation functor defined by $(X[1])^n = X^{n+1}$, then the data $(1')$ are objects X_{pq}^n of $\mathcal{B}$, the data $(2')$ are morphisms $X_{pq}^n \rightarrow X_{p'q'}^n$, the data $(3')$ are morphisms $X_{pq}^n \rightarrow X_{qr}^{n+1}$, and (c') becomes a long exact sequence

$$\cdots \rightarrow X_{pr}^n \rightarrow X_{pq}^n \xrightarrow{\partial} X_{qr}^{n+1} \rightarrow X_{pr}^{n+1} \rightarrow \cdots$$

Example 10. With the preceding notation, if F is a cohomological functor [6, §3.1] from $\mathcal{D}$ to $\mathcal{B}$, and if F^* is the functor from $\mathcal{D}$ to $\mathcal{A}$ defined by $F^n(X) = F(X[n])$, then F^* carries spectral objects of $\mathcal{D}$ to spectral objects of $\mathcal{A}$.

Remark 11. As in A.1, if X is a spectral object of $\mathcal{A}$, then for the morphisms $(2')$ to be isomorphisms for $p' \leq p \leq a$ and $q' = q$, and also for $b < q' \leq q$ and $p' = p$, it is necessary and sufficient that $X_{p,p+1} = 0$ for $p \notin [a, b]$. One then says that X has amplitude in $[a, b]$. In this case, $X_{-\infty, q}$, respectively $X_{p, \infty}$, denotes any $X_{p, q}$ with $p \leq a$, respectively $b < q$.

A.3

Let X be a spectral object of $\mathcal{A}$. For $p \leq q \leq r \leq s$, put

$$E(pqrs) := \text{Im}(X_{qs} \rightarrow X_{pr}). \quad (12)$$

The morphism $X_{qs} \rightarrow X_{pr}$ factors through X_{qr} , and this factorization gives a cross of exact sequences

$$\begin{array}{ccc} & X_{qr} & \\ X_{qs} \nearrow & & \searrow X_{pr} \\ & X_{qs}/X_{rs} & \end{array}$$

It follows that one also has

$$E(pqrs) = H(X_{qs} \rightarrow X_{qr} \rightarrow X_{pr}). \quad (13)$$

For $k \leq \ell \leq m \leq n \leq p \leq q$, the commutative diagram

$$\begin{array}{ccc} X_{\ell n} & \longrightarrow & X_{nq} \\ \downarrow & & \\ X_{km} & \longrightarrow & X_{mp} \end{array}$$

gives, after passing to images by the vertical morphisms (2'), a degree-one morphism

$$\partial: E(k\ell mn) \rightarrow E(mnpq).$$

For $k \leq \ell \leq m \leq n \leq p \leq q \leq r \leq s$, the composite

$$E(k\ell mn) \xrightarrow{\partial} E(mnpq) \xrightarrow{\partial} E(pqrs) \quad (14)$$

is zero, because the rows in the diagram

$$\begin{array}{ccccc} X_{\ell n} & \longrightarrow & X_{nq} & \longrightarrow & X_{qs} \\ \downarrow & & \downarrow & & \downarrow \\ X_{km} & \longrightarrow & X_{mp} & \longrightarrow & X_{pr} \end{array}$$

have zero composite (see A.2).

Construction. The cohomology of the sequence (14) is $E(\ell npr)$.

The cohomology of (14) is also the cohomology of

$$X_{\ell n} \xrightarrow{\partial} \text{Im}(X_{nq} \rightarrow X_{mp}) \xrightarrow{\partial} X_{pr}.$$

Consider the commutative diagram

$$\begin{array}{ccccc} & & X_{mn} & & \\ & \swarrow & & \searrow & \\ X_{\ell n} & & & & X_{nq} \\ & \searrow & & \swarrow & \\ & & X_{pq} & & X_{mp} \\ & & \searrow & \nearrow & \\ & & & X_{pr} & \end{array}$$

The desired cohomology is still that of

$$X_{\ell n} \rightarrow H(X_{mn} \rightarrow X_{np} \rightarrow X_{pq}) \rightarrow X_{pr},$$

which is equal to

$$H(X_{\ell n} \rightarrow X_{np} \rightarrow X_{pr}),$$

and this gives the required construction by (13).

A.4

For $r \geq 1$, put

$$E_r^p = E(p - r + 1, p, p + 1, p + r).$$

The morphism ∂ of A.3, for $(p - r + 1, p, p + 1, p + r, p + r + 1, p + 2r)$, is a degree-one map $d_r: E_r^p \rightarrow E_r^{p+r}$. One has $d_r^2 = 0$, and A.3 gives

$$E_{r+1}^p = H(E_r^{p-r} \rightarrow E_r^p \rightarrow E_r^{p+r});$$

the E_r^p form a spectral sequence.

If X has finite amplitude, say in an interval $[a, b]$ (A.2), this spectral sequence converges to $X_{-\infty, \infty}$; E_r^p is independent of r for sufficiently large r ($r > b - a$), and identifies with the associated graded of $X_{-\infty, \infty}$ filtered by the images of the $X_{p, \infty}$.

If this filtration is denoted by F , the exact sequence

$$X_{p, \infty} \rightarrow X_{-\infty, \infty} \rightarrow X_{-\infty, p}$$

indeed shows that

$$X_{p, \infty} \twoheadrightarrow F^p \quad \text{and} \quad X_{-\infty, \infty}/F^p \hookrightarrow X_{-\infty, p},$$

so that

$$F^p/F^{p+1} \sim \text{Im}(X_{p, \infty} \rightarrow X_{-\infty, p+1}) = E_{\infty}^p.$$

The spectral sequence constructed in this way starts from E_1 . In the case where $\mathcal{A}$ is the category of graded objects of $\mathcal{B}$, if one wants a spectral sequence starting from E_2 , it is enough to reindex:

$${}_2E_2^{pq} := ({}_1E_1^{-q})^{p+q}.$$

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